

Chapter 1 Assessment

Math 90-S61, Fall 2026, Jack Love

Last name, first name, ID#:

SOLUTIONS

Please read the following instructions carefully.

- You have until 12:25 PM to complete the assessment.
- Use the backs of pages for rough work/scratch paper. Show all relevant work neatly in the spaces provided, and box all final answers.
- You are welcome to ask me clarifying questions.

Here is a list of formulas you may find useful.

- The average rate of change of a function f on an interval $[a, b]$ is

$$AV_{[a,b]} = \frac{f(b) - f(a)}{b - a}$$

- An equation for a line with slope m containing point (x_0, y_0) is

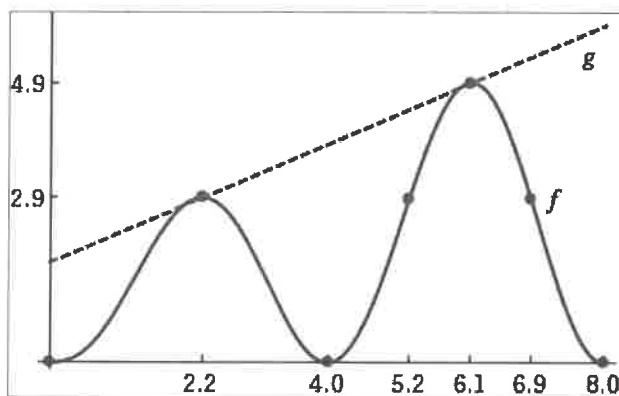
$$y = y_0 + m(x - x_0)$$

- A quadratic equation in standard form $q(x) = ax^2 + bx + c$ has

$$\text{zeros: } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad y\text{-intercept: } c \quad \text{vertex: } \left(-\frac{b}{2a}, q\left(-\frac{b}{2a}\right) \right)$$

- A quadratic equation in vertex form $q(x) = a(x - h)^2 + k$ has vertex (h, k) .
- A quadratic equation in factored form $q(x) = a(x - r)(x - s)$ has zeros $x = r$ and $x = s$.

1. (10 pts) The image shows the graphs of $f(x)$ and $g(x)$.



(a) (2 pts) Find $f(6.9) =$ 2.9

- (b) (2 pts) Fill in the blanks in each of the two points below to correctly complete the coordinates of two points on the graph of $g(x)$.

(2.2, 2.9)

(6.1, 4.9)

- (c) (2 pts) For what values of x is/are $f(x) = 0$? If there is more than one value, write your answer as a comma-separated list of numbers.

$x =$ 0, 4.0, 8.0

- (d) (2 pts) For what values of x is/are $f(x) = g(x)$? If there is more than one value, write your answer as a comma-separated list of numbers.

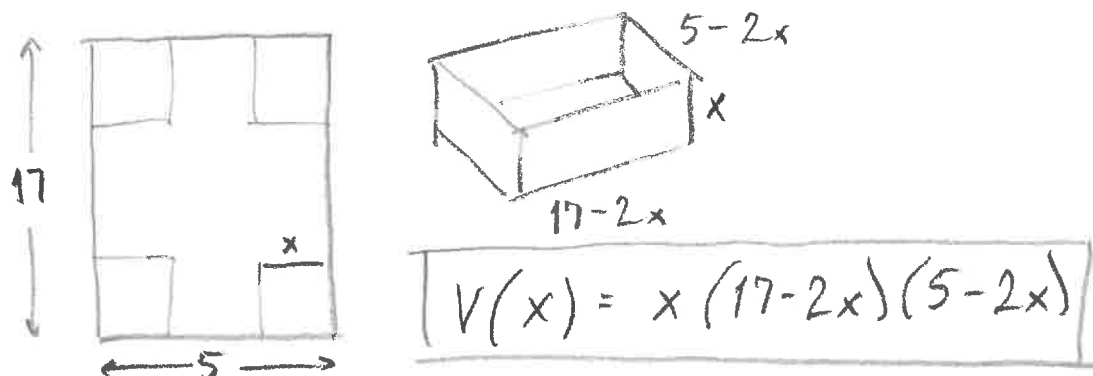
$x =$ 2.2, 6.1

- (e) (2 pts) What is the average rate of change of $f(x)$ on the interval $4.0 \leq x \leq 6.9$?

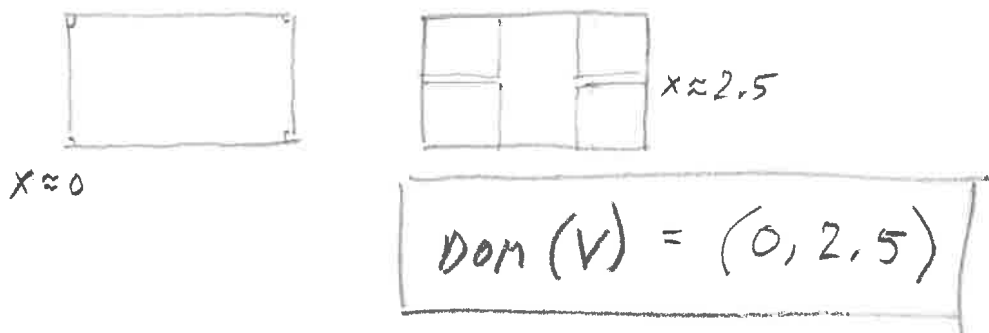
$$\begin{aligned} AV_{[4.0, 6.9]} &= \frac{f(6.9) - f(4.0)}{6.9 - 4.0} \\ &= \frac{2.9 - 0}{2.9} \\ &= \boxed{1} \end{aligned}$$

2. (8 pts) An open box is to be made from a flat piece of material 17 inches long and 5 inches wide by cutting equal squares of length x from the corners and folding up the sides.

(a) (6 pts) Write the volume V of the box as a function of x . Leave it as a product of factors, do not multiply out the factors.



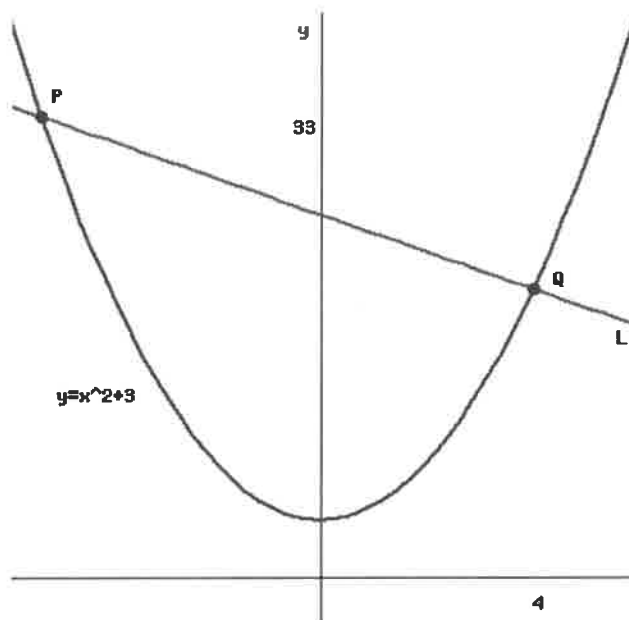
(b) (2 pts) Write the domain of the volume function V as an open interval in the form (a, b) .



3. (4 pts) Let $f(x) = 36 - x^2$. Compute the average rate of change on the interval $[1, 5]$. Write your answer as a positive or negative whole number.

$$\begin{aligned}
 AV_{[1,5]} &= \frac{f(5) - f(1)}{5 - 1} \\
 &= \frac{36 - 5^2 - (36 - 1^2)}{4} \\
 &= \frac{-25 + 1}{4} \\
 &= -6
 \end{aligned}$$

4. (10 pts) Consider the figure below. The x -coordinate of the point Q is 4, the y -coordinate of the point P is 33, and the parabola has equation $y = x^2 + 3$. Answer the following questions. Leave any radicals in radical form; do not approximate.



- (a) (4 pts) Determine the coordinates of the points P and Q .

$$\begin{aligned} P &= (-\sqrt{30}, 33) \\ Q &= (4, 19) \end{aligned}$$

$$P = (x, 33) \Rightarrow 33 = x^2 + 3 \Rightarrow x^2 = 30 \Rightarrow x = \pm\sqrt{30}$$

$$Q = (4, y) \Rightarrow y = 4^2 + 3 = 19$$

- (b) (4 pts) Use P and Q to determine the slope of the line L .

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{33 - 19}{-\sqrt{30} - 4} = \frac{14}{-\sqrt{30} - 4}$$

- (c) (2 pts) Use your answers from (a) and (b) to write an equation for L .

$$y = y_0 + m(x - x_0)$$

$$y = 33 + \frac{14}{-\sqrt{30} - 4} \cdot (x + \sqrt{30})$$

$$y = 19 + \frac{14}{-\sqrt{30} - 4} \cdot (x - 4)$$

OR

5. (8 pts) Consider the following quadratic function, whose graph is a parabola:

$$f(x) = x^2 + 5x - 14.$$

- (a) (6 pts) Determine the vertex of the parabola.

$$\text{VERTEX} = \left(-\frac{b}{2a}, f\left(-\frac{b}{2a}\right)\right), a=1, b=5, c=-14$$

$$-\frac{b}{2a} = -\frac{5}{2}, f\left(-\frac{5}{2}\right) = \left(-\frac{5}{2}\right)^2 + 5\left(-\frac{5}{2}\right) - 14$$

$$\boxed{\text{VERTEX} = \left(-\frac{5}{2}, -\frac{81}{4}\right)} \\ = (-2.5, -20.25)$$

$$= \frac{25}{4} - \frac{25}{2} - 14$$

$$= -\frac{81}{4}$$

- (b) (2 pts) Determine the y -intercept of the parabola.

$$f(0) = 0^2 + 5 \cdot 0 - 14 = \boxed{-14}$$

6. (6 pts) Find the zeros (if any) of the quadratic function

$$y = 3(x-5)^2 - 9.$$

Leave any radicals in radical form; do not approximate.

$$0 = 3(x-5)^2 - 9 \Rightarrow 9 = 3(x-5)^2 \Rightarrow 3 = (x-5)^2$$

$$\Rightarrow \pm\sqrt{3} = x-5 \Rightarrow \boxed{x = 5 \pm \sqrt{3}}$$

7. (12 pts) Given that $f(x) = 2x - 7$ and $g(x) = x + 5$, calculate the following.

(a) (3 pts) $(f \circ g)(3)$

$$f(g(3)) = f(8) = \boxed{9}$$

(b) (3 pts) $(g \circ f)(3)$

$$g(f(3)) = g(-1) = \boxed{4}$$

(c) (3 pts) $(f \circ f)(3)$

$$f(f(3)) = f(-1) = \boxed{-9}$$

(d) (3 pts) $(g \circ g)(3)$

$$g(g(3)) = g(8) = \boxed{13}$$

8. (8 pts) Let the functions f and g be given by the following table.

x	0	1	2	3	4
$f(x)$	4	6	3	6	1
$g(x)$	3	0	2	4	1

(a) (2 pts) Find $g^{-1}(0)$

$$\boxed{g^{-1}(0) = 1}$$

(b) (2 pts) Find $g(g^{-1}(2))$

$$\boxed{g(g^{-1}(2)) = 2}$$

(c) (4 pts) For what value(s) of x is $f(g(x)) = 6$?

$$f(g(x)) = 6 \Rightarrow g(x) = 1, 3$$

$$\Rightarrow \boxed{x = 0, 4}$$

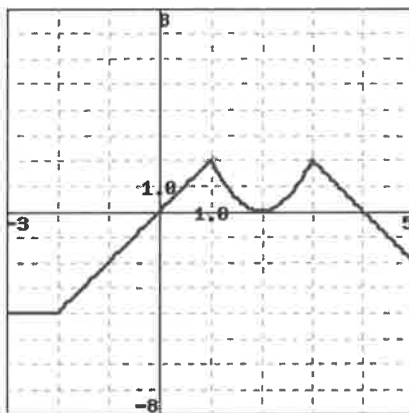
9. (4 pts) The cost (in dollars) of producing x air conditioners is $C = g(x) = 30x + 597$. Find a formula for the inverse function $g^{-1}(C)$.

$$g^{-1}(C) = \frac{C - 597}{30}$$

$$C = 30x + 597 \Rightarrow C - 597 = 30x$$

$$\Rightarrow x = \frac{C - 597}{30}$$

10. (12 pts) The figure below is the graph of a function $m(t)$.



Let $n(t) = m(t) + 2$, $k(t) = m(t + 1.5)$, $w(t) = m(t - 0.5) - 2.5$, and $p(t) = m(t - 1)$. Find the values of each of the following:

- (a) (3 pts) $p(2)$

$$p(2) = m(1) = \boxed{2}$$

- (b) (3 pts) $w(2.5)$

$$w(2.5) = m(2) - 2.5 = \boxed{-2.5}$$

- (c) (3 pts) $k(1.5)$

$$k(1.5) = m(3) = \boxed{2}$$

- (d) (3 pts) $n(-2.3)$

$$n(-2.3) = m(-2.3) + 2 = \boxed{-2}$$

11. (6 pts) The graph of $f(x)$ contains the point $(-5, 7)$. What point must be on each of the following transformed graphs?

(a) (2 pts) The graph of $f(x - 2)$ must contain the point

$(-3, 7)$

$\rightarrow$ 2 UNITS

(b) (2 pts) The graph of $f(x) + 3$ must contain the point

$(-5, 10)$

$\uparrow$ 3 UNITS

(c) (2 pts) The graph of $f(x + 5) - 4$ must contain the point

$(-10, 3)$

$\leftarrow$ 5 UNITS, $\downarrow$ 4 UNITS

12. (12 pts) For $f(x) = x^2 - 5$ and $g(x) = 4x + 1$, find:

(a) (4 pts) $(f \cdot g)(-3)$

$$f(-3) = 4, \quad g(-3) = -11. \quad (f \cdot g)(-3) = \boxed{-44}$$

(b) (4 pts) $(f - g)(2)$

$$f(2) = -1, \quad g(2) = 9. \quad (f - g)(2) = \boxed{-10}$$

(c) (4 pts) $\left(\frac{f}{g}\right)(0)$

$$f(0) = -5, \quad g(0) = 1. \quad \left(\frac{f}{g}\right)(0) = \boxed{-5}$$